

Mechanisms and Applications of the CRISPR-Cas9 Endonuclease System

A Review of Bacterial Adaptive Immunity and Genome Modification Pathways

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Abstract

The CRISPR-Cas9 system has transformed molecular genetics by providing a site-specific, programmable endonuclease for target genome modification. This review outlines bacterial CRISPR locus architecture, RNA-guided Cas9 cleavage mechanisms, and cellular double-strand break repair pathways.

1. Bacterial Adaptive Immunity Origins

Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) and CRISPR-associated (Cas) proteins were discovered as an adaptive immune defense in bacteria and archaea. Bacteria integrate short foreign DNA fragments (spacers) from invading bacteriophages into CRISPR genomic loci, forming a molecular memory used to degrade viral DNA upon secondary exposure.

2. Structure and Function of the Cas9 Endonuclease

The Type II CRISPR system from *Streptococcus pyogenes* utilizes a single endonuclease protein, Cas9, guided by a synthetic single guide RNA (sgRNA). The sgRNA contains a 20-nucleotide targeting sequence complementary to the target DNA. Binding requires an adjacent Protospacer Adjacent Motif (PAM, 5'-NGG-3'). Upon target site binding, Cas9's RuvC and HNH nuclease domains induce a blunt double-strand DNA break (DSB) 3 base pairs upstream of the PAM motif.

3. Post-Cleavage DNA Repair Pathways

Following Cas9-mediated double-strand break induction, eukaryotic cells repair the lesion using one of two endogenous pathways:

1. Non-Homologous End Joining (NHEJ): An error-prone repair mechanism operating throughout the cell cycle that frequently introduces insertions or deletions (indels), leading to frame-shift mutations and targeted gene knockouts. 2. Homology-Directed Repair (HDR): Operates during S/G2 phases, utilizing an exogenous donor DNA template to achieve precise nucleotide substitutions or gene insertions.

4. Therapeutic Prospects and Precision Engineering

CRISPR-Cas9 technology enables targeted gene therapy for monogenic disorders such as sickle cell disease and beta-thalassemia. Continued engineering of base editors and prime editors further enhances targeting precision while minimizing off-target cleavage events.

References

- [1] Jinek, M., Chylinski, K., Fonfara, I., Hauer, M., Doudna, J. A., & Charpentier, E. (2012). A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science*, 337(6096), 816-821.
- [2] Cong, L., et al. (2013). Multiplex genome engineering using CRISPR/Cas systems. *Science*, 339(6121), 819-823.
- [3] Doudna, J. A., & Charpentier, E. (2014). The new frontier of genome engineering with CRISPR-Cas9. *Science*, 346(6213), 1258096.